

**Instructions:** Homework is graded primarily on correctness and completion.

- Write your solutions on separate sheets of paper with your name written on each problem.
  - Include all key steps and explanations that justify your answers, unless it says otherwise.
  - Submit your written work as a .pdf file.
  - Attach all supporting files as a .zip file if applicable.
- 

## Homework Set 3

1.

For each linear homogeneous 2nd-order ODE with constant coefficients:

- find the characteristic equation,
- solve for the roots of the characteristic equation, and
- determine the general and specific solution of the ODE.

a)  $y'' - 2y' - 2y = 0$ ;  $y(0) = 1$ ,  $y'(0) = 1$

b)  $y'' = 6y' - 9y$ ;  $y(0) = 1$ ,  $y'(0) = 0$

c)  $y'' + y' + y = 0$ ;  $y(0) = 0$ ,  $y'(0) = 1$

d)  $y'' = -2y$ ;  $y(0) = 1$ ,  $y'(0) = 1$

2.

For each linear non-homogeneous 2nd-order ODE with constant coefficients:

- determine the homogeneous solution  $y_h$ ,
- find the particular solution  $y_p$  using either Undetermined Coefficients or Variation of Parameters, and
- write the general solution  $y = y_h + y_p$  and determine the specific solution.

Make sure to specify what method you used to find the particular solution.

a)  $y'' - 2y' + y = e^t$ ;  $y(0) = 1$ ,  $y'(0) = 0$

b)  $y'' + y = \cos(t)$ ;  $y(0) = 0$ ,  $y'(0) = 1$

c)  $y'' - 2y' + 2y = 2$ ;  $y(0) = 0$ ,  $y'(0) = 1$

d)  $y'' + y' - 2y = e^{-2t}$ ;  $y(0) = 1$ ,  $y'(0) = 0$

3.

Consider the linear 2nd-order ODE with constant coefficients:

$$y'' - y' = f(t).$$

Use either Undetermined Coefficients or Variation of Parameters to find the particular solution for each case of  $f(t)$ . Make sure to specify what method you used to find the particular solution.

a)  $f(t) = 0$

b)  $f(t) = 1$

c)  $f(t) = \frac{1}{1 + e^{-t}}$

d)  $f(t) = e^t (\sin(t) + 1)$

**4.**

Resonance occurs when a system is forced at its natural frequency, leading to a build-up of the amplitude of oscillation and energy. In other words, it occurs when the frequency of the non-homogeneous term matches the frequency of the homogeneous solution.

To illustrate resonance in its simplest form, consider the following linear non-homogeneous 2nd-order ODE with constant coefficients:

$$y'' + 9y = 3 \sin(\omega t); \quad y(0) = 1, \quad y'(0) = 0$$

where  $\omega$  is a real number.

- a) Determine the homogeneous solution.
- b) For each value of  $\omega$ , find the particular solution using either Undetermined Coefficients or Variation of Parameters. Make sure to specify what method you used to find the particular solution.
  - i.  $\omega = 1$
  - ii.  $\omega = 2$
  - iii.  $\omega = 3$
- c) Write the general solution and determine the specific solution for each value of  $\omega$  in Part (b).
- d) Use a graphing technology to plot the specific solutions determined in Part (c).
- e) Describe your observations of the behavior of solutions as  $\omega$  approaches the frequency of the homogeneous solution.

**5. Challenge Problem**

Consider the following linear non-homogeneous 2nd-order ODE with constant coefficients:

$$y'' + \alpha y' + \omega_0^2 y = \beta \cos(\omega t); \quad y(0) = 1, \quad y'(0) = 0$$

where  $\alpha$ ,  $\omega_0$  and  $\omega$  are real numbers.

- a) Let  $\alpha \neq 0$ . For each case below, determine the homogeneous solution.
  - i. Real roots, if  $\alpha > 2\omega_0$ .
  - ii. Repeated roots, if  $\alpha = 2\omega_0$ .
  - iii. Complex roots, if  $\alpha < 2\omega_0$ .
- b) Find the particular solution in two ways:
  - i. Using the method of Undetermined Coefficients, and
  - ii. Using the method of Variation of Parameters.

Then, compare the particular solutions obtained by each method.

- c) Write the general solution for each case in Part (a). Then, solve for the specific solution.
- d) Let  $\alpha = 0$ . Use your results in Part (c) to determine the general and specific solution for this case.
- e) Given the specific solution determined in Part (d) for the case when  $\alpha = 0$ , show what happens to the behavior of the solution as  $\omega$  approaches  $\omega_0$ .